

Gosford High School

Year 11

2007

Preliminary Higher School Certificate

Mathematics

Assessment Task 4

Time Allowed – 2.5 hours

Remember to start each new question on a new page

Students must answer questions using a blue/black pen and/or a sharpened B or HB pencil.

Approved scientific calculators may be used

Students need to be aware that

- * 'bald' answers may not gain full marks.
- * untidy and/or poorly organised solutions may not gain full marks.

Question 1 (12 marks)

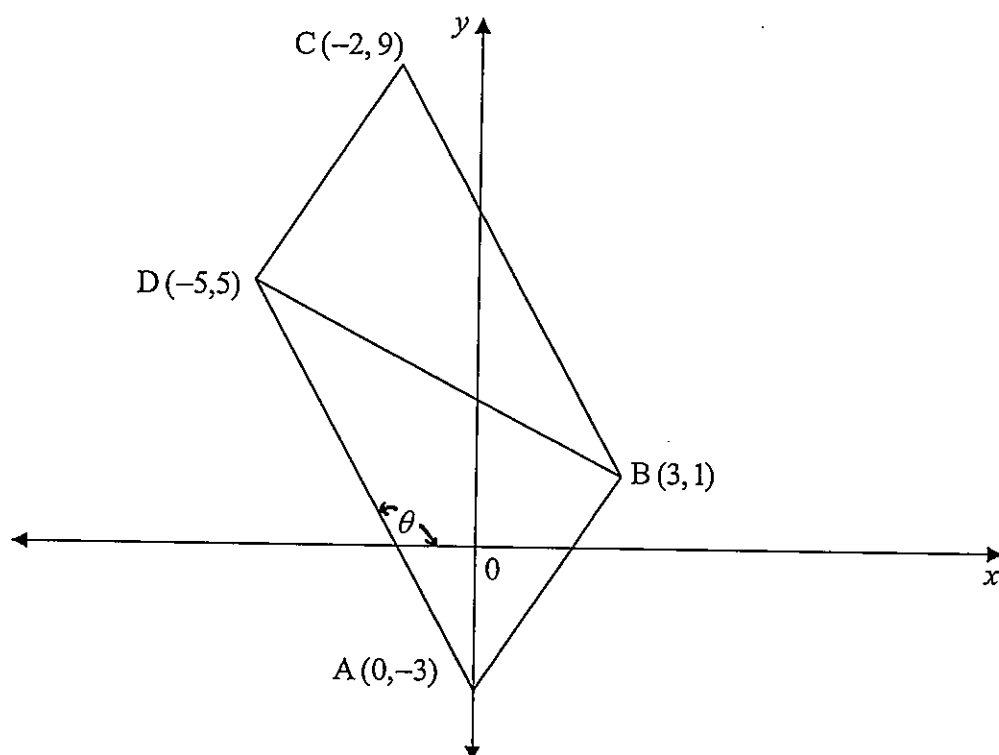
- (a) Factorise $x^3 + 27$ (1)
- (b) Write down the exact value of $\cos(210)^\circ$ (2)
- (c) Solve $2^x = \frac{1}{\sqrt{2}}$ (1)
- (d) Solve $|2x - 7| \leq 5$ (2)
- (e) Solve $\frac{3x}{5} - 4 = \frac{x}{2}$ (2)
- (f) In 2007 Council rates increased by $7\frac{1}{2}\%$. The new rate for a property is \$1735.
What was the old rate for this property. Give your answer correct to the nearest dollar. (2)
- (g) The speed of light is 299 725 km/h. Write this number correct to 3 significant figures. (1)
- (h) Find the value of $\sqrt[3]{456.7 \div 3\pi}$ correct to 1 decimal place. (1)

Question 2 (12 marks)

- (a) Solve the equations $3x - 2y = 7$ and $y = 3 - 2x$ simultaneously (2)
- (b) Solve $6x^2 - 7x - 20 = 0$ (2)
- (c) Find rational numbers a and m if $\frac{\sqrt{2}}{3\sqrt{2} - 2\sqrt{3}} = a + m\sqrt{6}$ (3)
- (d) Simplify $\frac{x+6}{x^2-4} - \frac{2}{x-2}$ (3)
- (e) Solve the equation $2x^2 = x + 5$ giving your answers in simplest surd form (2)

Question 3

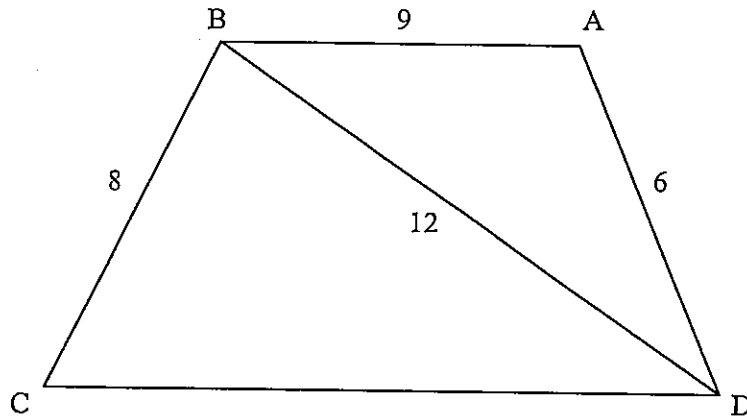
(12 marks)



- (a) Find the gradient of AD (1)
- (b) Find θ correct to the nearest degree (1)
- (c) Find the coordinates of M the midpoint of BD (1)
- (d) Prove that ABCD is a parallelogram (2)
- (e) Show that the equation of the line AB is $4x - 3y - 9 = 0$ (3)
- (f) Find the perpendicular distance from the point D to the line AB. (2)
- (g) Find the area of the quadrilateral ABCD (2)

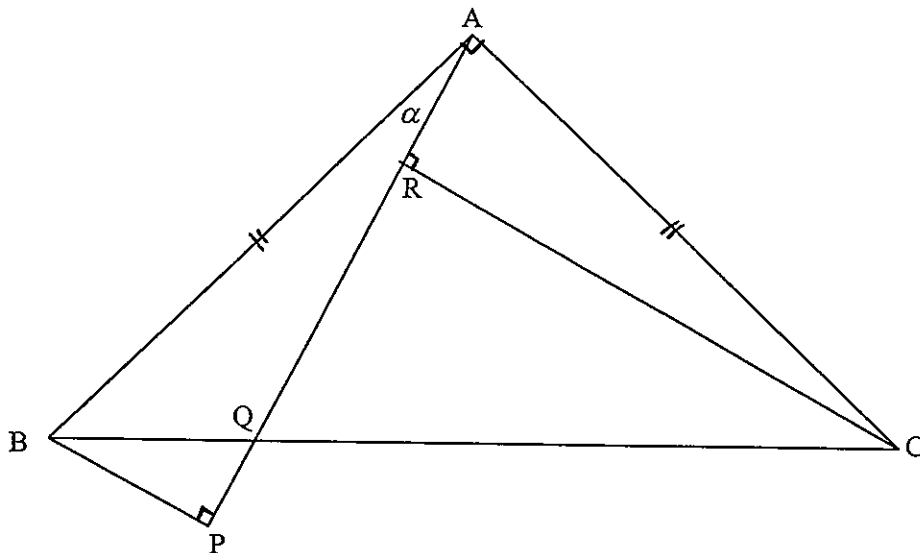
Question 4 (12 marks)

(a) In the diagram $\angle DAB = \angle CBD$



- (i) Prove triangles ABD and BDC are similar. (3)
- (ii) Find the length of CD (1)
- (iii) Prove that AB and CD are parallel (1)

(b)



In the diagram above $AB = AC$, $\angle BAC = \angle BPA = \angle CRA = 90^\circ$; $\angle BAP = \alpha$

Prove that

- (i) $\angle ACR = \alpha$ (1)
- (ii) triangles ABP and CAR are congruent (2)
- (iii) triangles BPQ and CRQ are similar (2)
- (iv) $\frac{PQ}{QR} = \frac{RA}{AP}$ (2)

Question 5

(12 marks)

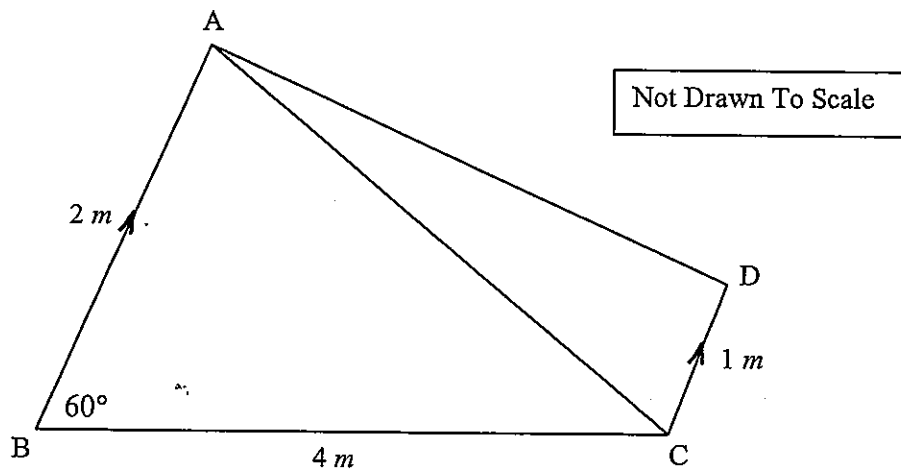
- (a) Shade the region satisfying the following inequations simultaneously $\begin{cases} y \leq 0 \\ y \leq 1 - x^2 \\ y > \left| \frac{x}{2} \right| \end{cases}$ (3)

- (b) In the diagram below ABCD is a quadrilateral in which AB is parallel DC, $\angle ABC = 60^\circ$, AB = 2 metres, BC = 4 metres and DC = 1 metre.

(i) Find the exact area of triangle ABC (2)

(ii) Find the exact length of AC (2)

(iii) Find the exact length of AD (3)



- (c) Evaluate $\lim_{x \rightarrow 3} \left[\frac{x^3 - 3x^2 + x - 3}{x - 3} \right]$ (2)

Question 6 (12 marks)

(a) Draw neat labelled sketches of each of the following functions:

(i) $y = (x - 2)^3$ (2)

(ii) $y = 1 + 2^x$ (2)

(iii) $y = 2x(4 - x)$ (2)

(b) State the domain and range of the function $y = \sqrt{9 + x^2}$ (2)

(c) (i) Find the value of $G(-2) + G(3)$ given that $G(x) = \begin{cases} x + 5 & \text{for } x < 0 \\ \sqrt{25 - x^2} & \text{for } x \geq 0 \end{cases}$ (1)

(ii) Is the function continuous at $x = 0$? (1)

(d) Write a quadratic equation with roots $3 + \sqrt{5}$ and $3 - \sqrt{5}$ (2)

Question 7 (12 marks)

(a) Find the derivatives of

(i) $7x^3 - 2x^2 + x - 8$ (1)

(ii) $y = \sqrt{4 - x^3}$ (2)

(b) Show that if $y = \frac{7x}{x^2 + 1}$ then $\frac{dy}{dx} = \frac{7(1 - x^2)}{(x^2 + 1)^2}$ (2)

(c) Find $f'(-1)$ if $f(x) = 2x(3 + 2x)^5$ (2)

(d) Find the equation of the normal to the curve $y = x^3 - x + 5$ at the point $(2, 11)$ on the curve. (2)

(e) Find the coordinates of the point on the curve $y = 3x^2 - 2x + 1$ where the tangent is parallel to the line $4x - y - 1 = 0$ (3)

Question 8**(12 marks)**

- (a) Simplify $\frac{1 - \cos^2 x}{\cos(90^\circ - x)}$ (2)
- (b) If $\tan \theta = \frac{5}{12}$ and $\cos \theta < 0$, find the exact value of $\operatorname{cosec} \theta$ (3)
- (c) Solve $\cos^2 \theta - 1 = 0$ for $0^\circ \leq \theta \leq 360^\circ$ (3)
- (d) (i) State the range of the function $y = \sin x$ (1)
(ii) Hence state the minimum value of the expression $7 + 6 \sin x$ (1)
- (e) Solve $\tan \theta = -\frac{1}{\sqrt{3}}$ for $0^\circ \leq \theta \leq 360^\circ$ (2)

Question 9**(12 marks)**

- (a) If α and β are the roots of the equation $2x^2 + 4x - 3 = 0$, what is the value of :
- (i) $\alpha + \beta$ (1)
- (ii) $\alpha\beta$ (1)
- (iii) $\alpha^2 + \beta^2$ (2)
- (b) Prove that $x^2 + (k-3)x - k$ has real roots for all values of k . (2)
- (c) $a(x-1)(x-2) + b(x-1) + c \equiv 2x^2 - 3x + 5$. Find a , b and c (3)
- (d) The quadratic equation $2x^2 - (m+2)x + m = 0$ has one root which is twice the other. Find the values of m . (3)

YR 11 MATHEMATICS PRELIMINARY FINAL
QUESTION 1 2007

a) $(x+3)(x^2-3x+9)$

b) $\cos 210^\circ = -\frac{\sqrt{3}}{2}$

c) $2^x = 2^{-\frac{1}{2}}$

$\therefore x = -\frac{1}{2}$

d) $-5 \leq 2x-7 \leq 5$

$2 \leq 2x \leq 12$

$1 \leq x \leq 6$

e) $\frac{3x}{5} - 4 = \frac{x}{2}$

$6x - 40 = 5x$

$x = 40$

f) $107\frac{1}{2}\%$ of Old Rate = \$1735

$\therefore \text{Old Rate} = \frac{\$1735}{1.075}$

$= \$1614$ (to nearest \$)

g) 300000 km/h

h) 3.6

QUESTION 2

a) $3x-2(3-2x) = 7$

$3x-6+4x = 7$

$7x = 13$

$x = \frac{13}{7}$

$x = 1\frac{6}{7}$

$y = 3-2\left(\frac{13}{7}\right)$

b) $(3x+4)(2x-5) = 0$
 $x = -\frac{4}{3}, \frac{5}{2}$

c) $\frac{\sqrt{2}}{3\sqrt{2}-2\sqrt{3}} \times \frac{(3\sqrt{2}+2\sqrt{3})}{(3\sqrt{2}+2\sqrt{3})}$

$= \frac{6+2\sqrt{6}}{18-12}$

$= \frac{6+2\sqrt{6}}{6}$

$= 1 + \frac{1}{3}\sqrt{6}$

$\therefore a = 1, m = \frac{1}{3}$

d) $\frac{x+6}{(x+2)(x-2)} - \frac{2}{x-2}$

$= \frac{x+6-2(x+2)}{(x+2)(x-2)}$

$= \frac{x+6-2x-4}{(x+2)(x-2)}$

$= \frac{2-x}{(x+2)(x-2)}$

$= \frac{-1}{x+2}$

e) $2x^2-x-5 = 0$

$x = \frac{1 \pm \sqrt{1-4(2)(-5)}}{4}$

$x = \frac{1 \pm \sqrt{41}}{4}$

QUESTION 3

a) $m_{AD} = \frac{y_2-y_1}{x_2-x_1}$

$= \frac{-3-5}{0+5}$

$= -\frac{8}{5}$

b) $\tan \theta = -\frac{8}{5}$

$\theta = \tan^{-1}\left(-\frac{8}{5}\right)$

$\theta = 122^\circ$

c) Midpoint of BD: $\left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right)$

$= \left(\frac{-5+3}{2}, \frac{5+1}{2}\right)$

$= \left(-\frac{2}{2}, \frac{6}{2}\right)$

$= (-1, 3)$

d) Midpoint of AC: $\left(\frac{-2+0}{2}, \frac{9-3}{2}\right)$

$= \left(-\frac{2}{2}, \frac{6}{2}\right)$

$= (-1, 3)$

$\therefore$ ABCD is a parallelogram

Since diagonals of ABCD

bisect each other

e) $m_{AB} = \frac{1+3}{3}$

$= \frac{4}{3}$

Equation of AB is

$y = \frac{4}{3}x - 3$

$3y = 4x - 9$

$4x - 3y - 9 = 0$

f) $d = \frac{|ax_1+by_1+c|}{\sqrt{a^2+b^2}}$ $b=-3$ $y_1=5$
 $c=-9$

$= \frac{|-20-15-9|}{\sqrt{16+9}}$

$= \frac{44}{5} \text{ units}$

g) $d_{AB} = \sqrt{(x_2-x_1)^2 + (y_2-y_1)^2}$

$= \sqrt{(3-0)^2 + (1+3)^2}$

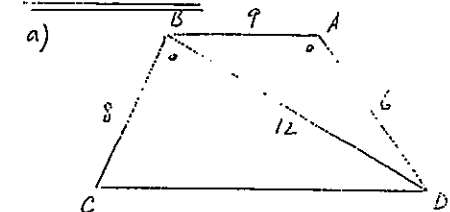
$= \sqrt{9+16}$

$= 5 \text{ units}$

Area ABCD = $5 \times \frac{44}{5}$

$= 44 \text{ sq units}$

QUESTION 4



i) Proof

In Δ 's ABD & BDC

$\hat{BAD} = \hat{CBD}$ (given)

$\frac{AB}{AD} = \frac{9}{6} = \frac{3}{2}$

$\frac{BD}{BC} = \frac{12}{8} = \frac{3}{2} = \frac{AB}{AD}$

$\therefore \Delta ABD \parallel \Delta BDC$ (one pair of sides in proportion and an included angle)

(ii) $\frac{CO}{BD} = \frac{BC}{AD}$
 $\frac{CO}{12} = \frac{8}{6}$
 $CO = 16$

(iii) $\hat{ABD} = \hat{BDC}$ (corresp. $\angle$'s of similar Δ 's in (i))

$\therefore AB \parallel CD$ (alternate $\angle$'s are equal)

b) $\hat{RAC} = 90^\circ - \alpha$ (adj. compl. $\angle$'s)

$\hat{ACK} = 180^\circ - [(90^\circ - \alpha) + 90^\circ]$

($\angle$ sum of ΔACK is 180°)

$\hat{ACK} = \alpha$

(ii) Proof

$\hat{BAP} = \hat{ACK} = \alpha$ (proven in (i))

$\hat{BPA} = \hat{ARC}$ (given right $\angle$'s)

$AB = AC$ (given)

$\therefore \Delta BAP \equiv \Delta ACR$ (AAS)

(iii) $\hat{BRA} = 180^\circ - \hat{CRA}$
 $= 180^\circ - 90^\circ$
 $= 90^\circ$

(adj. suppl. $\angle$'s)

$\therefore \hat{BRA} = \hat{BPA}$
 $= 90^\circ$

$\hat{BQP} = \hat{COR}$ (vert. opp. $\angle$'s)

$\therefore \Delta BPQ \parallel \Delta CRQ$

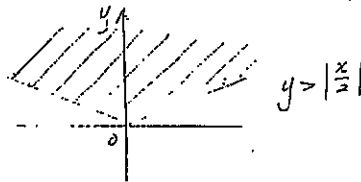
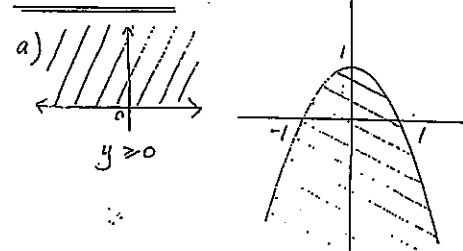
iv) $\frac{PB}{RQ} = \frac{BP}{RC}$

but $BP = AR$ and
 $RC = AP$

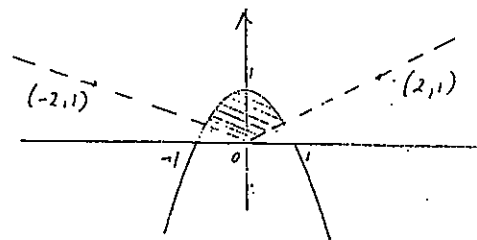
corresp. $\angle$'s of congruent Δ 's)

$\therefore \frac{PQ}{RR} = \frac{RA}{AP}$

QUESTION 5



Solution



b) (i) $A = \frac{1}{2} ab \sin C$
 $= \frac{1}{2} \times 2 \times 4 \times \sin 60^\circ$
 $= 4 \times \frac{\sqrt{3}}{2}$
 $= 2\sqrt{3}$ square units

(ii) $AC^2 = 2^2 + 4^2 - 2 \times 2 \times 4 \times \cos 60^\circ$
 $= 20 - 8$

(iii) Let $\hat{ACB} = \alpha$

$\therefore \frac{\sin \alpha}{2} = \frac{\sin 60^\circ}{2\sqrt{3}}$

$\sin \alpha = \frac{2 \times \frac{\sqrt{3}}{2}}{2\sqrt{3}}$

$= \frac{1}{2}$

$\alpha = 30^\circ$

$\therefore \hat{ACD} = 180^\circ - (30^\circ + 60^\circ)$

(co-int. $\angle$'s, $AB \parallel DC$)

$\hat{ACD} = 90^\circ$

$\therefore AD^2 = AC^2 + DC^2$

$= (2\sqrt{3})^2 + 1^2$

$= 12 + 1$

$= 13$

$\therefore AD = \sqrt{13}$ metres

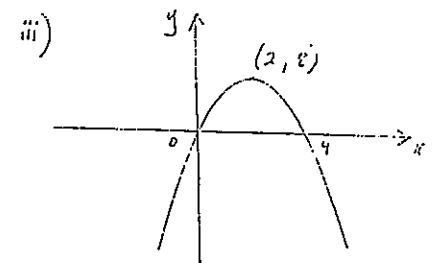
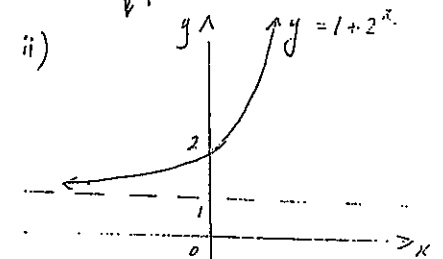
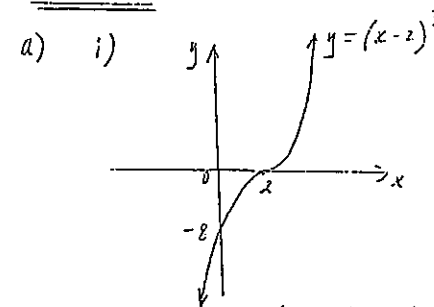
c) $\lim_{x \rightarrow 3} \left[\frac{x^2(x-3) + (x-3)}{(x-3)} \right]$

$= \lim_{x \rightarrow 3} \left[\frac{(x-3)(x^2+1)}{x-3} \right]$

$= \lim_{x \rightarrow 3} [x^2+1] \text{ for } x \neq 3$

$= 10$

Question 6



b) Domain x is any Real number

Range $y \geq 3$

c) i) $G(-2) + G(3) = 3 + 4$
 $= 7$

ii) Yes

d) $x^2 - (3+\sqrt{5}+3-\sqrt{5})x + (3+\sqrt{5})(3-\sqrt{5}) = 0$
 $x^2 - 6x + 4 = 0$

QUESTION 7

$$a) i) \frac{dy}{dx} = 21x^2 - 4x + 1$$

$$ii) y = (4 - x^3)^{\frac{1}{2}}$$

$$\frac{dy}{dx} = \frac{1}{2} (4 - x^3)^{-\frac{1}{2}} \times (-3x^2)$$

$$= \frac{-3x^2}{2\sqrt{4 - x^3}}$$

$$b) \frac{dy}{dx} = \frac{(x^2 + 1) \times 7 - 7x \times 2x}{(x^2 + 1)^2}$$

$$= \frac{7x^2 + 7 - 14x^2}{(x^2 + 1)^2}$$

$$= \frac{7 - 7x^2}{(x^2 + 1)^2}$$

$$= \frac{7(1 - x^2)}{(x^2 + 1)^2}$$

$$c) f'(x) = (3 + 2x)^5 \times 2 + 2x \times 5(3 + 2x)^4 \times 2$$

$$= 2(3 + 2x)^5 + 20x(3 + 2x)^4$$

$$f'(-1) = 2 \times 1 + (-20) \times 1$$

$$= 2 - 20$$

$$= -18$$

$$d) y = x^3 - x + 5$$

$$\frac{dy}{dx} = 3x^2 - 1$$

$$= 11 \text{ at } x = 2$$

When $x = 1$, the Normal is $y = -1$

Equation of Normal is

$$y - 11 = \frac{-1}{11}(x - 2)$$

$$11y - 121 = -x + 2$$

$$x + 11y - 123 = 0$$

$$e) y = 3x^2 - 2x + 1$$

$$\frac{dy}{dx} = 6x - 2$$

$$\therefore 6x - 2 = 4$$

$$6x = 6$$

$$x = 1$$

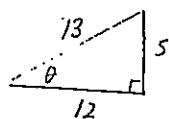
$\therefore$ Pt is $(1, 2)$

QUESTION 8

$$a) \frac{1 - \cos^2 x}{\cos(90^\circ - x)} = \frac{\sin^2 x}{\sin x}$$

$$= \sin x$$

b) θ lies in 3rd Quadrant.



$$\operatorname{cosec} \theta = \frac{1}{\sin \theta}$$

$$= \frac{1}{-\frac{5}{13}}$$

$$= -\frac{13}{5}$$

$$c) \cos^2 \theta = 1$$

$$\cos \theta = \pm 1$$

$$\therefore \theta = 0, 180^\circ, 360^\circ$$

$$d) i) -1 \leq y \leq 1$$

$$ii) \text{Minimum Value} = 7 + 6(-1)$$

$$= 1$$

$$e) \tan \theta = -\frac{1}{\sqrt{3}}$$

$$\theta = 150^\circ, 330^\circ$$

QUESTION 9

$$a) (i) \alpha + \beta = -\frac{b}{a} \quad (ii) \alpha\beta = \frac{c}{a}$$

$$= -\frac{4}{2} \quad = -\frac{3}{2}$$

$$= -2$$

$$(iii) \alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$$

$$= (-2)^2 - 2 \times \left(-\frac{3}{2}\right)$$

$$= 4 + 3$$

$$= 7$$

b) For Real Roots

$$b^2 - 4ac \geq 0$$

$$b^2 - 4ac = (k - 3)^2 - 4(1)(-k)$$

$$= k^2 - 6k + 9 + 4k$$

$$= k^2 - 2k + 9$$

$$= k^2 - 2k + 1 + 8$$

$\therefore b^2 - 4ac \geq 0$
Since $(k - 1)^2$ and 8 are both positive

c) True for $x = 1$

$$\therefore c = 2 - 3 + 5$$

$$c = 4$$

True for $x = 2$

$$\therefore b + c = 8 - 6 + 5$$

$$b + 4 = 7$$

$$b = 3$$

True for $x = 0$

$$\therefore 2a - b + c = 5$$

$$2a - 3 + 4 = 5$$

$$2a = 4$$

$$a = 2$$

d) Let α and 2α be the roots

$$\alpha + 2\alpha = \frac{m+2}{2} \quad 2\alpha \times \alpha = \frac{m}{2}$$

$$3\alpha = \frac{m+2}{2} \quad 2\alpha^2 = \frac{m}{2}$$

$$\alpha = \frac{m+2}{6} \quad \alpha^2 = \frac{m}{4}$$

$$\therefore \left(\frac{m+2}{6}\right)^2 = \frac{m}{4}$$

$$\frac{m^2 + 4m + 4}{36} = \frac{m}{4}$$

$$m^2 + 4m + 4 = 9m$$

$$m^2 - 5m + 4 = 0$$

$$(m - 4)(m - 1) = 0$$